Lab 2 - Configure Packet Filter Firewalls to Mitigate Attacks

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
Server1	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
Admin	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18
WebServer	NIC	192.168.2.3	255.255.255.0	192.168.2.1	N/A

Objectives

- Verify connectivity among devices before firewall configuration.
- Use ACLs to ensure remote access to the routers is available only from the Admin PC.
- Configure ACLs on R1 and R3 to mitigate attacks.
- Verify ACL functionality.

Background/Scenario

Access to routers R1 and R3 should only be permitted from the Admin PC. The Admin PC is also used for connectivity testing to Server1, which is a Server providing DNS, SMTP, FTP, and HTTPS services.

Standard operating procedure is to apply ACLs on edge routers to mitigate common threats based on source and destination IP address. In this activity, you will create ACLs on edge routers R1 and R3 to achieve this goal. You will then verify ACL functionality from internal and external hosts.

The routers have been pre-configured with the following:

- Enable password: **ciscoenpa55**
- Password for console: **ciscoconpa55**
- SSH logon username and password: **SSHadmin**/**ciscosshpa55**
- IP addressing
- Static routing

Part 1: Verify Basic Network Connectivity

Verify network connectivity prior to configuring the IP ACLs.

Step 1: From WebServer, verify connectivity to the Admin PC, Server1 and R1.

a. From the command prompt, ping the **Admin** PC (192.168.3.3) and **Server1** (192.168.1.3).

b. From the command prompt, establish an SSH session to **R1's** LAN interface (192.168.1.1) using username **SSHadmin** and password **ciscosshpa55**. When finished, exit the SSH session.

```
SERVER> ssh -l SSHadmin 192.168.1.1
```

QUESTION 1 Was the SSH connection from WebServer to R1 successful?

Step 2: From the Admin PC, verify connectivity to Server1, WebServer and R1.

a. From the command prompt, ping **Server1** (192.168.1.3) and **WebServer** (192.168.2.3).

b. From the command prompt, establish an SSH session to **R1's** LAN interface (192.168.1.1) using username **SSHadmin** and password **ciscosshpa55**. Close the SSH session when finished.

c. Open a web browser to **Server1** (192.168.1.3) to display the web page. Close the browser when done.

QUESTION 2 Was the Admin PC able to access the web page on Server1?

Before proceeding, go to the "Lab 2 - Configure Packet Filter Firewalls to Mitigate Attacks - QUESTIONS" quiz on the Brightspace page and enter your answers for questions 1 and 2. Leave the quiz open while you complete the rest of the lab sheet.

Part 2: Secure Access to Routers

Step 1: Configure ACL 10 to block all remote access to the routers except from the Admin PC.

a. From global configuration mode on **R1**, enter the following command to determine the first valid number for an extended access list.

```
R1(config)# access-list ?
```

QUESTION 3 Based on the output, what is the range of valid numbers for a standard ACL?

Before proceeding, return to the quiz on the Brightspace page and enter your answer for question 3. Leave the quiz open while you complete the rest of the lab sheet.

b. Use the **access-list** command to create a numbered ACL **10** on **R1** to block all remote access to the router except from the Admin PC.
Note: Since the ACL will automatically have an implicit "deny any" rule at the end of it by default, you are not expected to explicitly add a rule for this ACL to block traffic. In other words, your ACL just needs to define what traffic is allowed and nothing more.

c. Repeat Step 1b. on **R3**.

Step 2: Apply ACL 10 to ingress traffic on the VTY lines.

Use the **access-class** command to apply the access list to incoming traffic on the VTY lines on R1 and R3.

Step 3: Verify exclusive access from the Admin PC.

a. Establish an SSH session to 192.168.1.1 from the **Admin** PC.

b. Establish an SSH session to 192.168.1.1 from **WebServer**.

QUESTION 4 Which, if any, of the devices are still able to SSH to R1?

Before proceeding, return to the quiz on the Brightspace page and enter your answer for question 4. Leave the quiz open while you complete the rest of the lab sheet.

Part 3: Create a Numbered IP ACL 105 on R1

Create an IP ACL numbered 105 with the following rules:

- Permit any outside host to access DNS services on **Server1.**
- Permit any outside host to access SMTP services on **Server1.**
- Permit any outside host to access FTP services on **Server1.**
- Deny any outside host access to HTTPS services on **Server1.**
- Permit the **Admin** PC to access **R1's** WAN interface (S0/0/0) via SSH.

Note: Check Results will not show a correct configuration for an ACL until you have it fully configured correctly (with all of the rules in the correct order). See appendix of lab sheet for port numbers.

Step 1: Verify that the Admin PC can access the Server1 via HTTPS using the web browser.

Be sure to disable HTTP and enable HTTPS on **Server1** if needed.

Step 2: Configure ACL 105 to specifically permit and deny the specified traffic.

Use the **access-list** command to create a numbered IP ACL 105 based on the requirements listed above. See the appendix at the end for a list of the corresponding port numbers.

Step 3: Apply the ACL to interface S0/0/0.

Use the **ip access-group** command to apply the access list in the appropriate direction on interface S0/0/0.

Step 4: Verify that the Admin PC cannot access Server1 via HTTPS using the web browser.

Part 4: Create a Named IP ACL on R1

Step 1: Verify that Server1 cannot successfully ping the WebServer.

QUESTION 5 Why can Server1 not ping WebServer currently?

Before proceeding, return to the quiz on the Brightspace page and enter your answer for question 5.

Step 2: Configure a named extended ACL named ICMP_ACL to permit and deny the specified traffic.

Use the **ip access-list** command to create a named extended ACL using the name **ICMP_ACL**. The ACL should do the following (in this order):

- Permit ICMP echo replies from all sources to all destinations
- Permit destination unreachable messages from all sources to all destinations
- Deny all other incoming ICMP packets
- Permit all other types of traffic (from all sources to all destinations)

Hint: Refer to slide 48 of lecture 3 for help.

Step 3: Apply the ACL to interface G0/1.

Apply the access list in the appropriate direction on interface G0/1.

Note: In most real situations, these ACL rules would be applied to the external interface (i.e. S0/0/0). However, are applying it to the internal interface here for the purposes of this lab.

Part 5: Create a Numbered IP ACL 110 on R3

Only permit outgoing packets on **R3** that contain valid source addresses from inside the network and deny all others.

Step 1: Configure ACL 110 to permit all traffic from the inside network only.

Hint: Refer to slide 45 of lecture 3 for help. The ACL just needs to define what traffic is allowed and nothing more as all other traffic will be automatically denied by the implicit "deny ip any any" rule.

Step 2: Apply the ACL to interface G0/1.

Use the **ip access-group** command to apply the access list to incoming traffic on interface G0/1.

Part 6: Create a Numbered IP ACL 100 on R3

On **R3**, create the ACL to do the following (in this order):

- Block all packets containing a source IP address which is any of the three RFC 1918 private addresses (in ascending order)
- Block all packets containing a source IP address which is 127.0.0.0/8
- Block all packets containing a source IP address which is 255.255.255.255
- Permit all other traffic going from all other sources to all other destinations

Step 1: Configure ACL 100 to block all specified traffic from the outside network.

Use the **access-list** command to create a numbered IP ACL.

Hint: Refer to slide 45 of lecture 3 for help

Step 2: Apply the ACL to interface Serial 0/0/1.

Use the **ip access-group** command to apply the access list to incoming traffic on interface Serial 0/0/1.

Step 3: Confirm that the specified traffic entering interface Serial 0/0/1 is handled correctly.

From the Admin PC command prompt, ping WebServer. The ICMP echo replies are blocked by the ACL since they are sourced from the 192.168.0.0/16 address space.

If you have correctly configured all parts of the lab your activity score should now be showing as 100%. Show your activity file to your lab supervisor so they can verify your completion.

You have completed the lab - please submit the Brightspace quiz.

Appendix - Port Numbers

DNS - UDP 53
SMTP - TCP 25
FTP - TCP 21 (no need to include port 20)
HTTPS - TCP 443
SSH - TCP 22